

GDP Nowcasting Based on Daily Electricity Data and Financial Market Data

1st Wei Shang*

Center for Forecasting Science
Academy of Mathematics and Systems
Sciences, Chinese Academy of
Sciences
Beijing, China
shangwei@amss.ac.cn

2nd Xun Zhang

Center for Forecasting Science
Academy of Mathematics and Systems
Sciences, Chinese Academy of
Sciences
Beijing, China
zhangxun@amss.ac.cn

3rd Rui Tao

Center for Forecasting Science
Academy of Mathematics and Systems
Sciences, Chinese Academy of
Sciences
Beijing, China
taorui17@mails.ucas.ac.cn

4th Xiuzhenzi Wang

Center for Forecasting Science
Academy of Mathematics and Systems
Sciences, Chinese Academy of
Sciences
Beijing, China
wxzz97@163.com

5th *Lili Zhang

State Grid Energy Research Institute
Co., Ltd
Beijing, China
zhanglili@sgeri.sgcc.com.cn

Abstract—This paper proposed a macroeconomic nowcasting model that incorporates daily electricity data and financial market data, as well as other important economic statistical indicators. Mixed frequency dynamic factor model is employed to accommodate truncated data for predicting the quarterly growth rate of China's gross domestic product (GDP), and high-frequency data from the financial market and electricity market are included into the model to timely reflect the contribution of the latest economic activities to the quarter's economic growth, improving the timeliness of national economic forecasting. By dynamically updating electricity and economic statistics data, timely generation of quarterly GDP forecasts. This model can achieve nowcasting of the current quarter's GDP, which can more timely reflect the current state of economic activity and achieve higher prediction accuracy compared to existing models.

Keywords—GDP, Nowcasting, Dynamic factor model, Daily data processing

I. INTRODUCTION

Macroeconomic statistical data, such as gross domestic product (GDP), is relatively low frequency data which is often released months after actual economic activities occur. Currently, the preliminary calculation of China's annual GDP is generally released around January 20th of the following year, the final calculation of annual GDP is no later than the end of January of the following year, the preliminary calculation of quarterly GDP is generally released around the 15th day after the quarter, and the final calculation of quarterly GDP is generally released at the end of first month of the following quarter. The delay of GDP data releasement hinder the timely macroeconomy monitoring and policy decision making processes. There is a call for timely estimation of the GDP level of current period before the official data is published.

Nowcasting is a concept originated from meteorology, referring to the current state of the weather in detail and the prediction of changes that can be expected on a timescale of a few hours. Nowcasting in economics is the prediction of the very recent past, the present, and the very near future state of an economic indicator, using all available data at the moment. Economics nowcasting involves the technique of using the truncated data to forecast the most recent data of the target

economic indicator to be released. Development of economic modeling technology and big data applications makes it possible to do the GDP nowcasting using relatively higher frequency data, such as monthly economic data or even daily stock market data. The International Monetary Fund and Federal Reserve Bank had already released their GDP nowcasting in a timely manner.

Methods for GDP nowcasting currently available include: bridging model, distributed lag model, mixed frequency distributed lag model, mixed frequency sampling regression model, etc. The bridging model introduces monthly data into quarterly GDP forecasts, establishing a "bridge" between monthly statistical data and quarterly data, which was mainly used by early European central banks [1-3]. The bridging model contains few statistical variables and incomplete indicator information, but it has consistency and coherence. Koenig et al. proposed a distributed lag model [4]. Chen and Tsay improved the distributed lag model and proposed a generalized lag distribution model [5]. The MIDAS (Mixed Data Sampling) model is a widely used mixing prediction model proposed by Ghysels et al. [6]. Subsequently, many scholars proposed extended forms of the MIDAS model, such as the MS MIDAS model [7] and the cointegration MIDAS model [8]. In addition, there is the current prediction model MF-VAR for estimating joint endogenous variables [9]. Due to the limitations of the data dimensions of regression models, the above models are usually based on a few important variables in the national economic accounts, such as investment, consumption, foreign trade from a demand perspective, or the primary, secondary, and tertiary industries from a production perspective. The economic information contained in the models is incomplete and is also limited by the lag time of these monthly data releases.

Factor analysis is a statistical method used in multivariate statistical analysis to solve dimensionality reduction. Its idea is to use a few common factors to explain the complex relationships between multiple observation variables, in order to simplify the observation system and achieve the purpose of dimensionality reduction. From the early 20th century to the 1960s, factor analysis was mainly based on the "traditional factor model" proposed by Charles Spearman. This model was a static factor model, and its application in the economic field was limited to analyzing the structural characteristics of the

economic system, without the ability to analyze the dynamic characteristics of the macroeconomic system, such as reflecting the delayed and long-term sustained effects in economic activities or relationships, also known as the "lag effect". With the continuous development of model estimation techniques such as Kalman filtering and maximum likelihood, many assumptions of factor models have been continuously relaxed. Traditional factor models have been expanded to the Dynamic Factor Model (DFM) and further expanded to allow for mixing and truncation of data, that is to conquer the situation that the latest sample time of the data is inconsistent, and the data is incomplete. DFM is capable of incorporate massive and high-frequency data, thus more aspect of the economic and social system can be taken into consideration for the forecasting task. It is gradually becoming a major method in GDP nowcasting.

The mixed frequency dynamic factor model is currently a GDP nowcasting method being used by the Federal Reserve Bank of the United States, the European Central Bank, and other international institutions. These models typically include 30-150 economic and financial data. However, these models mainly conform to the characteristics of the European and American economies. For predicting China's GDP, it is necessary to select a more suitable, leading, and representative indicator set based on the characteristics of the Chinese economy, and design a nowcasting model. Wang Xia et al. [10], on the basis of collecting China's real-time fragmented tail dataset, adopted a mixed dynamic factor model to effectively solve the data problems that need to be faced in nowcasting, including mixing indicators, fragmented tail features Periodic missing of data, etc. Zheng Tingguo [11] selected financial variables as predictive factors based on real-time data and used the Mixed Data Sampling (MIDAS) model to short-term predict GDP growth rate, achieving good predictive results. Zhang Wei et al. [12] used a dynamic factor model to construct a high-frequency public opinion index, expanded existing low-frequency (up to monthly) prediction models, and proposed a monitoring system for nowcasting of China's macroeconomic aggregate. Power big data has a wide coverage, timely and stable updates, and can comprehensively and truly reflect the macroeconomic operation. Peng et al. [13] found that in GDP prediction, electricity data has better nowcasting performance compared to consumption, investment, and exports. Currently, there is a lack of models that effectively introduce daily power data into China's GDP nowcasting.

II. METHODOLOGY

A. Nowcasting with updated data

The solution of using the most updated data in GDP nowcasting includes updating and processing the real time data as well as the nowcasting model. Economic Calendar is essential in understanding the data update timing. Chinese Economic calendar is as shown in Fig. 1. Survey data, including PMI can be obtained right after the end of the month, that is in the 2nd to 4th day of the next month. Most financial data series will be available around the 11th to 20th day of the next month. Some Trading data and profit data will be available almost at the end of next month due to the sophisticated data collection and validation processes.

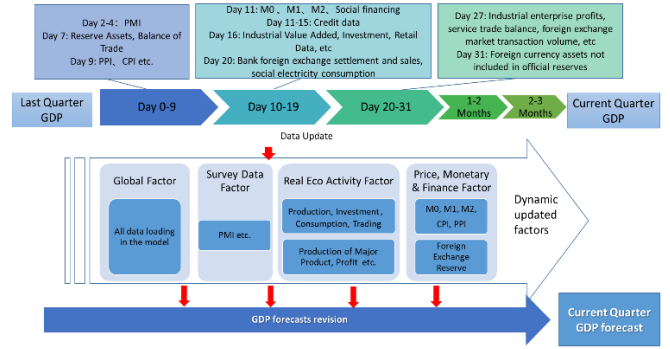


Fig. 1. Chinese Economic Calendar

Five steps are designed accordingly to construct a data update and modeling mechanism, as shown in Fig: (1) selecting indicators and setting model factors; (2) Update data according to the economic calendar and supplement the data; (3) Mix training set data/real-time data for frequency processing; (4) Estimating/updating dynamic factor models; (5) Update real-time forecasts of GDP.

After the model is established, when more real time data is available, new data need to be processed and the DFM need to be updated, and then GDP nowcasting result can be updated as shown in the dashed path in Fig 2.

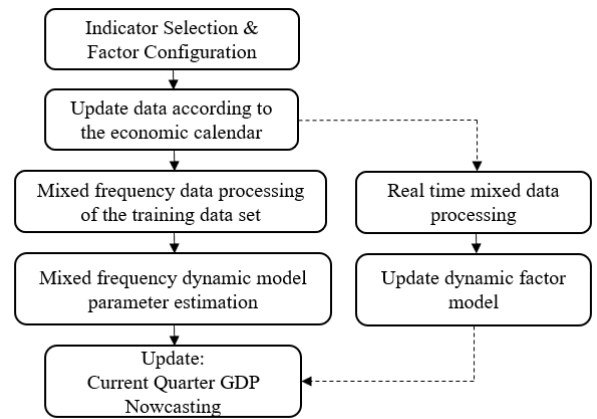


Fig. 2. Nowcasting framework

B. Economic variable selection

For an economic forecasting task, when select independent variables three criteria need to be consider: theoretical relevance, empirical correlations, and data availability. For nowcasting model data availability criteria can be loosen somehow, since the model is capability of dealing with truncated data. Selected indicators need to be grouped into several factors to capture different economic aspects.

In order to overcome the time lag and limited information set in existing macroeconomic forecasting, and provide more timely and accurate macroeconomic forecasting results, electricity data and financial market data are to be included in GDP nowcasting model together with key economic data. There are four categories of data to be considered as the candidate pool of economic variable for GDP nowcasting:

(1) Real economic activity data: including production, investment, consumption, import and export, main industrial product output, industrial enterprise profits, etc.;

(2) Monetary and financial data: including price indices such as CPI, PPI, import and export price index, as well as Shanghai Stock Exchange trading volume, Shanghai Stock Exchange index, and money supply indicators;

(3) Survey data: mainly Purchasing Manager's Index (PMI) and its components;

(4) Electricity data: including electricity generation, industrial electricity consumption, overall social electricity consumption, mining industry electricity consumption, secondary industry electricity consumption.

C. Dynamic factor model setting

The dynamic factor model assumes that each observable variable is influenced by two factors: on the one hand, there are several common factors (common factors) that affect all variables, and these common factors constitute the common components of the variable system; On the other hand, there are unique influencing factors for each observable variable (special factors, also known as "random perturbations", hereinafter collectively referred to as "random perturbations"), which are important factors that distinguish each observable variable and constitute the random components of the variable system. The entire variable system can be seen as a sum of common and random components.

Assuming $y_t = [y_{1,t}, y_{2,t}, \dots, y_{n,t}]'$, $t=1, \dots, T$ is the normalized n -dimensional time series vector, where $y_{1,t}$ is the target variable. Establish a mixing dynamic factor model as follows:

$$y_t = \Lambda f_t + \epsilon_t \quad (1)$$

$$f_t = A_1 f_{t-1} + A_2 f_{t-2} + \dots + A_p f_{t-p} + v_t \quad (2)$$

$$v_t \sim i. d. N(0, Q)$$

Among them, f_t is the common factor vector of $r * 1'$, $\epsilon_t = [\epsilon_{1,t}, \epsilon_{2,t}, \dots, \epsilon_{n,t}]'$ is a heterogeneous vector independent of f_t . Λ is the factor load matrix with $n * r$ dimensions, let $\chi_t = \Lambda f_t$ be the common component. ϵ_t satisfies a normal distribution and the cross-section is uncorrelated. Assuming a common factor f_t at the same time T satisfies the balanced VAR (Vector autoregression) process, $A_1, \dots, A_p$ is the $r * r$ matrix of the autoregressive coefficients.

D. Mixed frequency data processing

The model data consists of monthly indicators and daily indicators. The monthly indicators need to be updated and output daily based on the economic calendar (update date). Under each generation date, the output is the monthly value of the monthly indicator in the previous month. If this generation date is later than the update date of the monthly indicator, it will be output; otherwise, it will not be output. Daily indicators are divided into electricity daily indicators and non-electricity daily indicators. The cumulative value from the beginning of the month in which electricity indicators are generated to the current day, and the cumulative average value from the beginning of the month in which non electricity indicators are generated to the current day. The values of electricity indicators and non-electricity indicators on the n th day can be calculated from the daily data from the 1st to the n th day of the month.

Mixing processing of initial data for non-electric daily indicators:

$$Xn = \frac{1}{n} (\sum_{i=1}^n x_i) = \frac{x_1 + x_2 + \dots + x_n}{n} \quad (3)$$

Mixing processing of daily electricity data:

$$Xn = \sum_{i=1}^n x_i \quad (4)$$

E. Parameter estimation

The process of estimating parameters adopts the expectation maximization method proposed by Dempster, Laird, and Rubin, using the EM algorithm to perform parameter estimation in two alternating steps [14]. Note $l(Y, F; \theta)$ is the joint logarithmic likelihood function of T and f , where $Y = [y_1, \dots, y_T]$, $F = [f_1, \dots, f_T]$, $t=1, \dots, T$. Parameters in the EM algorithm can be estimated as following:

E-step: The expectation of the logarithmic likelihood condition of the data is calculated using the estimation from the previous iteration, $\theta(j)$

$$L(\theta, \theta(j)) = E_{\theta(j)}[l(Y, F; \theta) | \Omega_T] \quad (5)$$

M-step: By adjusting the θ To re estimate parameters by maximizing the expected logarithmic likelihood:

$$\theta(j+1) = \underset{\theta}{\operatorname{argmax}} L(\theta, \theta(j)) \quad (6)$$

Repeat the above steps until convergence and obtain the estimated values of the parameters θ^* Please refer to Marta Ba for the specific estimation process n Bura and Michele Modugno [15].

F. Update of nowcasting results

Whenever the indicators in the economic calendar are updated, update the model input data and run the model again to update the estimated values of common factors, and then calculate the real-time predicted value of the target variable GDP.

When updating data at time s ,

$$y_T^s = [y_{1,T}, y_{2,T}, \dots, y_{n,T}]' = E[y_T | \Omega_s, \hat{\theta}] \quad (7)$$

Compared to the previous day, when updating the data at time $s-1$, the target variable y at time T . Nowcasting of $y_{1,T}^s$ The predicted adjustment range for $y_{1,T}^s$ is:

$$\Delta y_{1,T}^s = y_{1,T}^s - y_{1,T}^{s-1} \quad (8)$$

The proposed approach is capable of incorporate with data updated with various frequency at different time point so that it can fundamentally improve the timeliness of prediction. Unlike existing GDP prediction technologies, the proposed nowcasting model can be updated daily, with the daily updated financial market data and electricity data frequency, which is the highest frequency in the proposed model.

Although the model can be updated daily, updating the model and the nowcasting everyday may not the best choice in terms of nowcasting performance. On one hand the daily data source is limited to financial market data and electricity dispatch data. Daily electricity data is obtained from the central allocation monitoring, and cannot be divided into industries. Comparing with the large group of monthly updated data, calculate the model everyday may not bring much improvement, since most of the cases only minor daily information is added to the model. On the other hand the daily data contains more noise which is not reliable to be used solely

without other monthly meaningful economic data. Therefore, considering the pattern of Chinese economic calendar, this paper use 10th, 20th and last day of month as data updating time point, to leverage the information timeliness and the calculation efficiency.

However, since the dynamic factor model, mixed frequency data processing and nowcasting result update approaches is independent of input data structures, if more daily data is available, the proposed approach can be adapted and incorporate more daily updated data. In such circumstances, more update time points or even daily update can be adopted for GDP nowcasting tasks.

III. EMPIRICAL RESULTS

A. Data Source

To empirically validate the proposed GDP nowcasting approach, an experiment of predicting current quarter GDP is designed. The data sample period is from January 2013 to June 2023, corresponding to GDP data from Q1 2013 to Q2 2023. According to data availability, the model testing samples were from Q1 2021 to Q2 2023. Update the quarterly GDP forecast three times a month based on data availability, on the 10th, 20th, and 30th, respectively. Therefore, the test sample includes a total of 12 data points.

Economic data and financial market data are obtained from the Wind Database, which is one of the major economic and financial provider. Daily electricity consumption data represented by the is from the State Grid Corporation of China.

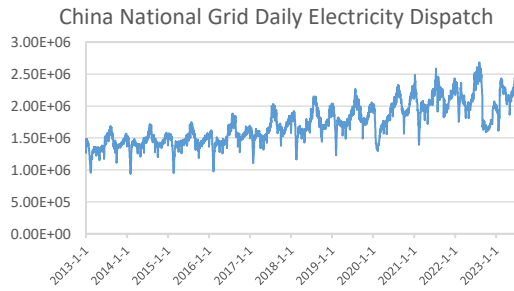


Fig. 3. China National Grid Daily Electricity Dispatch

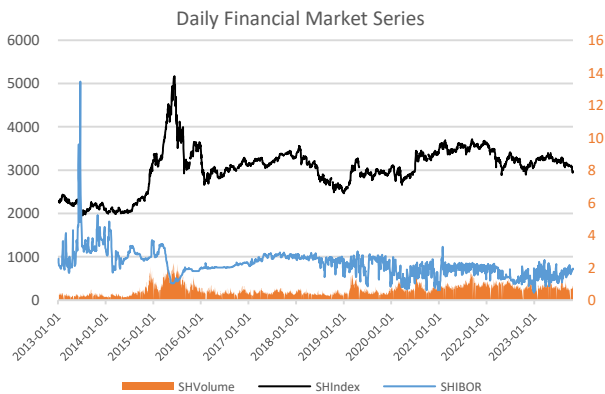


Fig. 4. Financial Market Data

B. Variable Selection

Based on the economic calendar, indicator significance, and statistical significance, 47 indicators are selected into the core GDP nowcasting model. The Chinese GDP is a quarterly indicator and the rest being monthly or daily indicators. The model includes four indicators that can be updated daily, namely the trading volume of the Shanghai Composite Index, the closing price of the Shanghai Composite Index, the Shanghai overnight lending rate, and the electricity consumption. The daily electricity consumption here is represented by national grid dispatch amount. The model sets factor loadings, assuming that all variables are affected by a single economic cycle factor. In addition, to characterize the correlation of similar indicators in the indicator sector, the model introduces three local common factors: real economic activity, price and monetary finance, and survey data. These three common factors have loadings on the corresponding specific economic indicator set based on the division in Table I.

TABLE I. INDICATORS SELECTED FOR CHINESE GDP NOWCASTING

	Indicators	Freq
Survey		
1	Caixin Manufacturing PMI: Business Activity Index	M
2	Caixin Manufacturing PMI	M
3	PMI: New Order	M
4	PMI	M
Real Economic Activities		
5	US: Export Amount: Current month YOY	M
6	United States: Import amount: Current Month YOY	M
7	Actual amount of foreign investment used: Foreign direct investment: year-on-year in the current month	M
8	Import amount: General trade: Current Month YOY	M
9	Export amount: General trade: Current Month YOY	M
10	Import amount: Current Month YOY	M
11	Export amount: Current Month YOY	M
12	Electricity consumption: Manufacturing industry: YOY Current Month	M
13	Electricity consumption: Industrial: YOY Current Month	M
14	Total social electricity consumption: YOY Current Month	M
15	Electricity consumption: Mining industry: YOY Current Month	M
16	Whole society Electricity consumption: secondary industry: year-on-year in the current month	M
17	Industrial added value: YOY Current Month	M
18	Completed amount of new fixed assets investment: Cumulative YOY	M
19	Completed investment in real estate development: Cumulative YOY	M
20	Completed amount of fixed assets investment: cumulative year-on-year	M
21	Total retail sales of consumer goods: YOY Current Month	M
22	Production: pig iron: YOY Current Month	M
23	Production: Automobile: YoY for the month	M

	Indicators	Freq
24	Industrial enterprise: finished product inventory: cumulative year-on-year	M
25	Industrial enterprise: Total profit: Cumulative YOY	M
26	Production: crude steel: YOY Current Month	M
27	Production: Steel: YOY Current Month	M
28	GDP: constant price: QoQ	Q
29	Electricity consumption for grid dispatch: Cumulative Value	D
Price, Monetary and Finance		
30	Import Price Index (HS2): Total Index	M
31	Export Price Index (HS2): Total Index	M
32	Official reserve assets: foreign exchange reserves: YOY	M
33	CPI: YOY Current Month	M
34	CPI:Food: YOY Current Month	M
35	CPI:Service: YOY Current Month	M
36	CPI:Consumption-Goods: YOY Current Month	M
37	CPI:Non-Food: YOY Current Month	M
38	PPI:Daily necessities: YOY Current Month	M
39	PPI: Production materials: Raw materials industry: YOY Current Month	M
40	PPI: Production materials: Raw materials industry: YOY Current Month	M
41	PPI: All industrial products: YOY Current Month	M
42	M0: YOY	M
43	M1: YOY	M
44	M2: YOY	M
45	Shanghai Interbank Offered Overnight Rate: Daily Average	D
46	Daily average trading volume of Shanghai Composite Index: YOY	D
47	Closing price of the Shanghai Composite Index: daily average	D

C. Data Processing and Preparation

For China's monthly economic data, the release time of each indicator is different. Taking the data released in August 2021 as an example, July CPI: year-on-year and July PPI: year-on-year (%) were released on August 9 (the next month), and July industrial added value, fixed assets investment and total retail sales of social consumer goods were released on August 16 (the next month). As a popular source of economic early signals, survey data, represented by manufacturing PMI data were released at the beginning of the month, that is around August 2.

In order to valid the proposed nowcasting model, it is essential to simulate the daily updated data as in practice. That is in each point of forecasting, data availability should be strictly follows the schedule as indicated in the economic calendar. Therefore a mechanism is design to update the data daily according to the economic calendar (update date), so that all data can be updated to the latest statistical values that can be obtained at the current time point.

And after that, data of different frequency need to be merged. Daily electricity consumption data need to be transformed to the cumulative value from the beginning of the

month to the current day of the generation date, while non power indicators, such as Shanghai Composite Index trading volume, Shanghai Composite Index closing price, Shanghai overnight lending rate etc., need to transform to the average value from the beginning of the month to the current day of the generation date.

When all the data of different frequency is prepared, the data file of a certain date point is ready. For each date, such data file containing all the indicator data available at the moment indicated by that date is to be generated. And a group of data file is generated, so that they are to be used to estimate the nowcasting model.

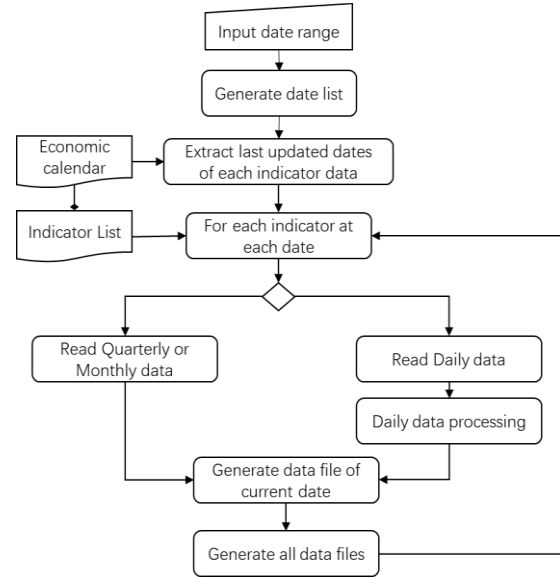


Fig. 5. Data Processing and Preparation

D. Model Estimation and Factor Loadings

EM algorithm is used to estimate the mixing dynamic factor model. After calculation, taking the last quarter of 2019 as an example, in monthly sequence factor loading matrix, the total profit of industrial enterprises has the maximum global factor loading of 0.2567, the pig iron has the maximum real economy factor loading of 0.3447, and non-food CPI has the maximum factor loading of 0.4513. Factor loadings of daily electricity data and financial data are shown in Table II.

TABLE II. PART OF THE FACTOR LOADINGS MATRICES

	Indicator Name	Global	Survey	Real	Price
1	Caixin Manufacturing PMI	0.1496	0.4523	0	0
2	Industrial enterprise: Total profit: Cumulative YOY	0.2567	0	-0.1688	0
3	PPI: Production materials: Raw materials industry: YOY Current Month	0.1759	0	0	0.1995
4	Industrial added value: YOY Current Month	0.1843	0	0.0698	0
5	Electricity consumption: Manufacturing industry: YOY Current Month	0.2256	0	0.1434	0
6	Electricity consumption: Industrial: YOY Current Month	0.2008	0	0.205	0
7	Total social electricity consumption: YOY Current Month	0.1994	0	0.1999	0

	Indicator Name	Global	Survey	Real	Price
8	Electricity consumption: Mining industry: YOY Current Month	0.2185	0	0.0815	0
9	Whole society Electricity consumption: secondary industry: year-on-year in the current month	0.203	0	0.1999	0
10	Electricity consumption for grid dispatch: Cumulative Value	0.0323	0	0.0342	0
11	Shanghai Interbank Offered Overnight Rate: Daily Average	-0.1321	0	0	-0.0385
12	Daily average trading volume of Shanghai Composite Index: YOY	-0.0468	0	0	-0.0589
13	Closing price of the Shanghai Composite Index: daily average	-0.1321	0	0	-0.0385

Daily electricity data and financial data reveal lower factor loading comparing with major economic indicators, such as the profit and price. It might be caused by the large fluctuation of these daily high frequency series. These daily data may serve as complimentary information source for GDP nowcasting although their contribution is not as much as traditional economic indicators.

Common factor of the whole sample data range is shown together with Real GDP and standardized series of all selected variables in Fig. 6. The common factor reveals similar fluctuation pattern with the real GDP series, which means the proposed model with selected variables is capable of capture the dynamic of GDP in the propose time period.

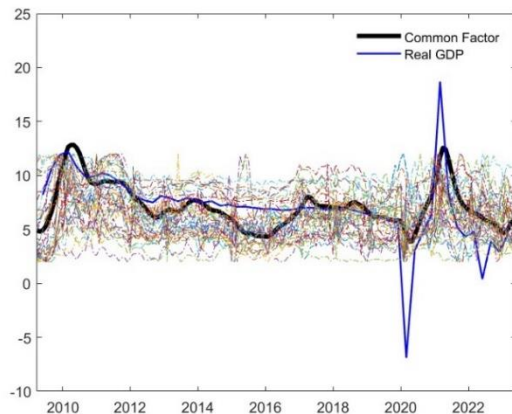


Fig. 6. Common factor and all economic variables in the model

E. Nowcasting performance

To further validate the predictive performance of the proposed nowcasting model, we conducted a rolling motion prediction progress evaluation. Considering the drastic impact brought by the Covid-19 pandemic, the testing period is designed starting before the pandemic, specifically from Jan. 2019 to the end of our sample data.

Three time points are used per month: 10th, 20th and 28/30/31th day, to distinguish the contribution of each updated data item. Both nowcasting of the current month and

forecasting of the next month are tested. The experiment is conducted as follow:

(1) On January 10, 2019, based on all available data as of January 10, 2019, a dynamic factor model was estimated. Based on this dynamic factor model, GDP for the first quarter of 2019 (at this time, the data was only updated to December 31, 2018) and GDP for the second quarter of 2019 were predicted.

(2) On January 20, 2019, obtain the latest data and re estimate the dynamic factor model to predict GDP for the first quarter and second quarter of 2019; And based on the model estimated on January 10, 2019, the GDP forecast for the second quarter of 2019 was calculated by substituting it into the new dataset on January 20, 2019.

(3) Repeat the above process until June 30th, 2023.

Taking the year 2019 as an example, nowcasting results for the year 2019 are shown in Table III. Since the Covid-19 pandemic cause a sharp drop in GDP in the first quarter of 2020, the impact is not included in the model, the predicted results of GDP for the first quarter of 2020 were not included in the evaluation. The RMSE of GDP nowcasting for current quarter by the model with electricity data is 0.305, while the RMSE predicted by the model without electricity data is 0.322; The prediction of GDP for the next quarter and the RMSEs of the two models were 0.622 and 0.672, respectively. Electricity data helps to improve the accuracy of current GDP prediction. If the dynamic factor model is re estimated based on new data as well as the predicted GDP data in a rolling manner, the model's predicted RMSE for the next quarter's GDP will decrease from 0.622 to 0.451 (including electricity data), or from 0.672 to 0.492 (excluding electricity data). Rolling update models can improve prediction accuracy.

TABLE III. NOWCASTING AND FORECASTING PERFORMANCE

Year 2019	Actual GDP	With Electricity			Without Electricity		
		Now	Forecast	Rolling Forecast	Now	Forecast	Rolling Forecast
Jan	6.3	6.578	6.741	/	6.524	6.705	/
Feb		6.254	6.347	6.344	6.344	6.46	6.45
Mar		6.521	6.773	6.765	6.463	6.638	6.637
Apr	6	6.549	6.706	6.559	6.573	6.758	6.597
May		6.577	6.771	6.762	6.623	6.839	6.825
Jun		6.483	6.585	6.583	6.559	6.702	6.699
Jul	5.9	6.024	6.252	6.059	6.093	6.389	6.141
Aug		5.798	6.002	6.002	5.926	6.175	6.18
Sep		5.85	6.107	6.111	5.965	6.251	6.25
Oct	5.8	6.025	6.21	6.03	6.027	6.223	6.052
Nov		5.881	6.028	6.035	5.88	6.031	6.037
Dec		5.994	6.277	6.28	5.999	6.278	6.28
RMSE		0.305	0.622	0.451	0.322	0.672	0.494
MAPE(%)		4.07	9.74	8.64	4.137	10.945	9.918

Nowcasting results for the year 2021 to 2023 are shown in Table IV. Average RMSE of GDP nowcasting by the model with electricity data is 4.54, and by the model without

electricity data is 4.69, nowcasting accuracy is increased by 3.4%. Along with the recover from the pandemic, nowcasting error RMSE is decreased from 6.36 to 3.05 and 0.56.

The daily electricity data, however, does not reveal consistent contribution to forecasting accuracy increase. In the year 2021, nowcasting RMSE is large due to the pandemic impact, the electricity data contribute to a 0.40 decrease in RMSE. While in the year of 2022, such contribution is not supported. The electricity data does not help to increase the nowcasting performance. In the year 2023, with the recovery of Chinese economy, the GDP nowcasting accuracy of the first two quarters increases. And the electronic data contribute to a 0.03 decrease in forecasting error.

TABLE IV. NOWCASTING PERFORMANCE ACROSS YEARS

Mon	2021		2022		2023	
	<i>Ele</i>	<i>No Ele</i>	<i>Ele</i>	<i>No Ele</i>	<i>Ele</i>	<i>No Ele</i>
Jan	8.10	7.42	5.68	5.86	3.40	3.63
Feb	9.38	8.10	5.38	5.59	3.06	3.45
Mar	9.93	6.59	6.20	5.92	4.18	4.07
Apr	15.84	14.31	6.05	5.43	6.08	5.72
May	14.83	14.23	5.52	5.19	6.17	5.45
Jun	14.55	14.05	5.61	5.49	/	/
Jul	10.23	10.28	4.57	4.34	/	/
Aug	9.13	9.20	4.27	3.87	/	/
Sep	8.41	8.22	4.38	4.02	/	/
Oct	7.22	6.35	5.93	6.06	/	/
Nov	6.65	5.45	5.62	5.64	/	/
Dec	6.63	5.56	5.40	5.46	/	/
RMSE	6.36	6.76	3.05	2.91	0.56	0.53
RMSE Decrease	0.4	/	-0.14	/	0.03	/
Over all RMSE	With Ele		No Ele		Improvement in Accu.	
	4.54		4.69		3.4%	

F. Contribution of updated data

To further validate the predictive performance of the proposed nowcasting model, we conducted a rolling motion prediction progress evaluation. Considering the drastic impact brought by the Covid-19 pandemic, the testing period is designed starting before the pandemic, specifically from Jan. 2019 to the end of our sample data.

TABLE V. NOWCASTING PERFORMANCE

Year 2021.	Actual GDP	10th		20th		28/30th	
		<i>Ele</i>	<i>No Ele</i>	<i>Ele</i>	<i>No Ele</i>	<i>Ele</i>	<i>No Ele</i>
Jan	18.7	8.23	7.30	8.09	7.37	8.10	7.42
Feb		9.44	8.13	9.41	8.12	9.38	8.10
Mar		9.48	8.10	9.90	6.56	9.93	6.59
Apr	8.3	15.43	14.12	15.85	14.18	15.84	14.31

Year 2021.	Actual GDP	10th		20th		28/30th	
		<i>Ele</i>	<i>No Ele</i>	<i>Ele</i>	<i>No Ele</i>	<i>Ele</i>	<i>No Ele</i>
May	5.2	15.84	14.84	14.84	14.27	14.83	14.23
Jun		14.93	14.53	14.59	14.09	14.55	14.05
Jul		10.69	11.10	10.32	10.35	10.23	10.28
Aug		10.01	10.16	9.20	9.28	9.13	9.20
Sep		8.64	8.90	8.44	8.24	8.41	8.22
Oct	4.3	7.61	7.74	7.30	6.41	7.22	6.35
Nov		6.92	6.40	6.74	5.59	6.65	5.45
Dec		6.51	5.48	6.64	5.50	6.63	5.56
RMSE		6.58	6.83	6.38	6.78	6.36	6.76
RMSE Decrease		/	/	0.2	0.05	0.02	0.02
Over all RMSE	With Ele		No Ele		Improvement in Accu.		
	6.44		6.79		5%		

G. Discussions

Theoretically, every meaningful information adding on a forecasting model will increase its forecasting performance, since new information will usually decrease the uncertainty and increase the forecasting accuracy. However, in practice, no forecasting model is perfectly designed with all necessary variables, due to the limitation of data availability and knowledge limit as well as the existence of unobservable and unexpected influential factors.

According to empirical results, the proposed nowcasting model incorporating daily electricity data and financial data does reveals improvement in both nowcasting and forecasting in average. GDP nowcasting accuracy increase along with more data is updated. Electricity data, especially may increase the forecasting accuracy, since it is a direct indicator associate with production and consumption activities. However, in special cases of pandemic burst and recovering period, the impact of electricity data on the model nowcasting accuracy is not stable. Complex situations such as all government measurements yet to be understood in modelling the GDP growth. Contribution of electricity data in such unusual circumstances is not consist in all time frames. However, general improvement of nowcasting and forecasting performance is achieved.

IV. CONCLUSIONS

This paper proposed a GDP nowcasting modeling approach incorporating daily electronic data and financial market data. Based on macroeconomic statistical data, survey data, and electricity data, a current macroeconomic forecasting model was constructed. Based on indicator correlation analysis, indicator importance, and availability screening, a mixed frequency dynamic factor model containing 47 variables and 4 factors was ultimately determined as the benchmark core model for real-time forecasting.

In the study, the factor load matrix and other coefficients of the nowcasting model were calculated and estimated. Taking the nowcasting of GDP in the fourth quarter of 2019 as an example, the update and adjustment process of nowcasting accompanied by data flow updates was introduced in detail. Through a comparative study of the prediction progress of nowcasting models with or without power data,

we found that: (1) Electricity data helps improve the accuracy of GDP nowcasting, (2) Rolling update models can improve the forecasting accuracy. (3) With the time approaching in each month, add on information increase the forecasting accuracy.

Through empirical analysis, the proposed nowcasting model can fully explore the economic information contained in macroeconomic data and electricity data. The model's prediction error for the current quarter's GDP growth rate is less than 5%. And the electricity data may contribute to 3.4% decrease in nowcasting error in general even during the pandemic recovering period. The proposed approaches is independent of country and data availability. It can be hopefully implemented for other countries with different data availability.

Future work may be designed using modeling techniques other than DFM. Graph Neural Network and other machine learning models could be an alternative choice in building the nowcasting model. And further investigations of the applicability of the proposed data mixing mechanisms and nowcasting model for countries other than China are also required to further valid the generalizability of the proposed approaches.

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REFERENCES

- [1] A. Baffigi, R. Golinelli, G. Parigi, "Bridge models to forecast the Euro area GDP". *International Journal of Forecasting*, 2004, vol. 20, no.3, pp. 447-460. DOI:10.1016/S0169-2070(03)00067-0.
- [2] F. Sédillot and N. Pain, "Indicator models of real GDP growth in selected OECD countries," *Economics Department working papers* no. 364 OECD, https://www.oecd-ilibrary.org/economics/indicator-models-of-real-gdp-growth-in-selected-oecd-countries_275257320252.
- [3] K. Barhoumi, O. Darné, L. Ferrara, B. Pluyaude, "Monthly GDP forecasting using bridge models: application for the French economy," *Bulletin of Economic Research*, 2012, vol. 64, s1, pp. s53-s70.
- [4] E.F. Koenig, S. Dolmas, J. Piger, "The use and abuse of real-time data in economic forecasting" *Review of Economics and Statistics*, 2003, vol 85, no.3, pp. 618-628.
- [5] Y. Chen, W.J. Tsay, "Forecasting commodity prices with mixed-frequency data: An OLS-based generalized ADL approach," *IEAS Working Paper*, 11-A001, 2011.
- [6] E. Ghysels, P. Santa-Clara, R. Valkanov. "The MIDAS touch: Mixed data sampling regression models". UC Los Angeles: Finance. <http://escholarship.org/uc/item/9mf223rs>, 2004.
- [7] P. Guérin P, M. Marcellino, "Markov-switching MIDAS models," *Journal of Business & Economic Statistics*, 2013, 31(1), pp. 45-56.
- [8] J.I. Miller, "Mixed-frequency cointegrating regressions with parsimonious distributed lag structures," *Journal of Financial Econometrics*, 2013, 12(no), pp. 584-614. DOI: 10.1093/jfinec/nbt010.
- [9] M. Modugno, "Now-casting inflation using high frequency data," *International Journal of Forecasting*, 2013, vol 29, no. 4, pp. 664-675, <http://dx.doi.org/10.1016/j.bbr.2011.03.031>.
- [10] X. Wang, N. Si, T. Song, "Nowcasting China's Quarterly GDP Using Mixed-Frequency Data", *Journal of Financial Research*, 2021, No.494(08), pp. 22-41.
- [11] T.G. Zheng, Y.H. Shang, "Mixed-Frequency Based Forecasting China's GDP with Financial Indices", *Journal of Financial Research*, 2013, No.399(09), pp.16-29.
- [12] W. Zhang, J.F. Tian, C. Cao, "Macroeconomics aggregation nowcasting based on mixed frequency big data," *Macroeconomics*, 2020(02), pp.15-29
- [13] F. Peng, G.Q. Peng, Y.R. Qi, T.T. Liu, X.L. Zhou, "Nowcasting of China's industrial added value based on electric power big data" *Telecommunications Science*, 2021, 37(7), pp.115-125.
- [14] A.P. Dempster, Laird N M, Rubin D B. "Maximum likelihood from incomplete data via the EM algorithm," *Journal of the royal statistical society: series B (methodological)*, 1977, 39(1): 1-22.
- [15] M. Bańbura, M. Modugno, "Maximum Likelihood Estimation of Large Factor Model on Datasets with Arbitrary Pattern of Missing Data," *Working Paper Series 1189*, European Central Bank, 2010.